

# Khanpanion: claims and sources ledger

All sources below were checked on September 10, 2026 unless stated otherwise. "Verified" describes the source statement, not BACKTRACK's effectiveness. Proposed designs and targets are labeled separately.

## Competition requirements

**Claim:** KEIC is a specific Enactus Philippines track with meaningful Khan use central to the challenge. **Status:** verified from the official page in the browser. **Safe wording:** "Prepared for KEIC 2026." **Avoid:** importing contradictory rules from the general early-stage track. Source: <https://enactus.ph/2026-national-competition/khan-academy-challenge>

**Claim:** Application Khan integration weight is 25%, other five criteria 15% each. Finals allocate 15% each to six criteria plus 10% pitch/Q&A. **Status:** verified on the same specific page. **Avoid:** claiming the finals Khan weight is also 25%.

**Claim:** Application period August 30–September 18; nationals October 9–10 at DLSU Manila; winning implementation November 1–March 31. **Status:** verified. **Limit:** the inherited September 18, 11:59 PM PHT time was not independently displayed in the specific page/form inspection. Use September 18 as the verified deadline and confirm the time.

**Claim:** Faculty adviser plus three students; one team per campus; no member/adviser overlap with the separate early-stage track. **Status:** verified. Institutional authorization still requires the team's confirmation.

**Claim:** Seven narratives within 300 words, supporting PDF within 15 slides. **Status:** verified in the live Typeform welcome and fields. **Caveat:** executive field has max\_length 300 despite a 300-word description. Source: <https://form.typeform.com/to/bJw9THj3>. Public definition read at <https://form.typeform.com/forms/bJw9THj3>. The short executive fallback is prepared for the apparent character constraint.

**Claim:** KEIC is terminal and does not confer 2026 World Cup representation. **Status:** verified on specific page and form welcome. **Conflict:** general form legal boilerplate mentions 2027 World Cup policy. Do not promise World Cup eligibility; seek organizer clarification if relevant.

**Claim:** Major awards split team awards and implementation grants; champion implementation grant is PHP 100,000. **Status:** verified on official KEIC page. **Avoid:** treating the full PHP 200,000 champion prize as an implementation budget or secured funding.

## Khan resources and integration

**Claim:** Khan has lessons, exercises, teacher reports, and assignment-score CSV exports. **Status:** verified official reporting guide. **Safe wording:** “Use authorized reports available to the partner.” **Avoid:** “All real-time learner events flow automatically into BACKTRACK.” Source: <https://support.khanacademy.org/hc/en-us/articles/360031129891-What-reporting-options-are-available-on-Khan-Academy-for-teachers-to-track-student-performance>

**Claim:** Current public build embeds official Khan videos and links to matched practice. **Status:** implementation claim, verify by browser playback and link checks before delivery. No public result API or school roster integration is implemented. Khan opens and self-reported practice are distinct from BACKTRACK answers.

**Claim:** Khan materials and brand have usage conditions. **Status:** official help article checked. **Safe wording:** credit Khan, preserve logo appearance, use official hosted videos and links, keep independent project attribution. **Avoid:** implying endorsement or charging specifically for access to free Khan content. Source: <https://support.khanacademy.org/hc/en-us/articles/202262954-Can-I-use-Khan-Academy-s-videos-name-materials-links-in-my-project>

**Claim:** Khan Academy Philippines has local curriculum/program activity. **Status:** official website checked; no BACKTRACK partnership verified. **Safe wording:** “Seek alignment and supported opportunities.” **Avoid:** claiming BACKTRACK is DepEd approved or already part of a Khan Philippines program. Source: <https://khanacademyphilippines.org/>

## Chemistry and physics coverage (added 11 September 2026)

**Claim:** Khanpanion supports nine destinations across mathematics, chemistry and physics. **Status:** implementation claim, verified in the browser on 11 September 2026 : nine destinations build and route, and a full physics journey (goal miss to prerequisite to original lab to two fresh unassisted checks to reviewer item with a future due date to Rematch) passes in the acceptance suite. **Safe wording:** “nine destinations across three subjects.” **Avoid:** “full chemistry and physics coverage,” “a science curriculum,” or any grade-level mapping.

**Claim:** Every generated chemistry and physics answer is correct. **Status:** verified by test, not by assertion. `tests/science.test.mjs` recomputes every formula mass from the published atomic-mass table, counts atoms on both sides of every generated equation, and checks every kinematics,  $F = ma$  and unit-conversion item against an independent computation. **Safe wording:** “answers are checked against an independently recomputed key in the test suite.” **Avoid:** claiming external curriculum review that has not happened.

**Claim:** Relative atomic masses are the IUPAC abridged standard atomic weights. **Status:** authored from the published abridged values and held in one module with the rounding policy beside them; **not** re-checked against the IUPAC source during this change. **Safe wording:** “a single reviewed constants table with a stated rounding policy.” **Avoid:** citing a specific IUPAC publication year in learner-facing or judge-facing copy until someone has checked the table against the source. This is a live item for the science reviewer.

**Claim:** Science steps connect to matched Khan Academy practice. **Status:** NOT true for the science-specific steps and must not be claimed. No chemistry or physics URL could be opened and title-checked by hand on this machine, so none shipped; those steps are marked unmatched and the interface says so. **Safe wording:** “physics routes reach verified Khan mathematics practice through their prerequisites; chemistry and physics steps have no matched Khan resource yet, and Khanpanion says so.” **Avoid:** any wording that implies Khan science content is integrated. See docs/THIRD\_PARTY\_MATERIALS.md.

**Claim:** The curriculum framing suits a particular Philippine grade level. **Status:** NOT established. The research is explicit that a generic “Grade 9 = X” mapping can mislead under the MATATAG transition. **Safe wording:** name the destination, not the grade. **Avoid:** asserting a grade level. Unmatched teacher topics stay visibly unmatched.

## Competitor overlap

**IXL:** continuously updated diagnostic recommendations. Verified product statement. <https://www.ixl.com/diagnostic> . Avoid invention-of-adaptive-learning claims.

**ALEKS:** knowledge-state assessment and custom objectives with prerequisites. Verified documentation. [https://www.aleks.com/about\\_aleks/index](https://www.aleks.com/about_aleks/index) and <https://www.mheducation.com/unitas/school/explore/sites/aleks/customizing-custom-objectives.pdf> . Avoid saying competitors do not understand current objectives.

**Khan Learning Paths:** documented requirements include a district partnership, MAP Growth and Clever. Guide updated June 28, 2026. <https://support.khanacademy.org/hc/en-us/articles/38442199334285-What-are-the-technical-requirements-for-implementing-Learning-Paths> . Avoid implying Khan lacks personalization.

**Gemini:** June 25, 2026 study notebooks describe goal-based diagnostics and adaptive lessons. <https://blog.google/innovation-and-ai/products/gemini-app/gemini-study-notebooks/> . Avoid “AI only helps if you already know what to ask.”

**Khanmigo:** established tutoring support. <https://www.khanmigo.ai/> . Avoid presenting it as a static content directory.

## Learning and motivation

**Retrieval and spacing:** supported learning principles, not an estimate of BACKTRACK’s effect. Sources: IES 2007 practice guide <https://ies.ed.gov/ncee/wwc/practiceguide/1> and Carpenter, Pan & Butler 2022 <https://doi.org/10.1038/s44159-022-00089-1> . Safe wording: “The design includes fresh and delayed checks.” Avoid guaranteed retention or a fabricated percentage gain.

**Confidence assessment:** Foster’s trial found no positive attainment difference. It used a different scoring intervention. <https://doi.org/10.1007/s10763-021-10207-9> . Safe wording: “Confidence is routing metadata to test.” Avoid treating it as evidence that BACKTRACK’s routing works.

**Autonomy and competence:** Self-Determination Theory informs choice and visible progress. <https://selfdeterminationtheory.org/topics/application-education/> . Safe wording: “Voluntary continuation is a hypothesis.” Avoid “science proves this mechanic is addictive.”

**Teaching at the Right Level:** an important precedent for targeting instruction to demonstrated level; its primary-school context and delivery model differ from this proposal. <https://www.povertyactionlab.org/case-study/teaching-right-level-improve-learning> . Avoid transferring its outcomes to a secondary mathematics app.

**Philippine remediation context:** DepEd Order 010, s. 2026 concerns the 2026 Summer Remediation Programs. <https://www.deped.gov.ph/2026/04/29/april-28-2026-do-010-s-2026-guidelines-for-the-implementation-of-2026-summer-remediation-programs/> . Safe wording: remediation is an active policy concern. Avoid saying prerequisite gaps explain the entire education crisis or that this order approves BACKTRACK.

## BACKTRACK-specific facts

**Team:** University of the Philippines Manila, coach Justin Mesias, students Matthew Labrador, Paul Recio, Harry Gomez. **Source:** Matthew’s explicit message in this work session. **Limit:** formal adviser eligibility, academic/contact fields, and partner-school status are not established by those names.

**UP Manila endorsement:** confirmed by Matthew in this session when requesting the official UP Manila logo on every slide. This is a team-supplied fact. Institutional endorsement does not establish school enrollment, a completed pilot, funding, or learning outcomes.

**Pilot reach:** 80–120 learners across two or three cohorts. **Status:** planning target conditional on access. No learners enrolled or outcomes claimed.

**Budget:** PHP 100,000 scenario. **Status:** planning allocation, not quotes or secured funds. Full actual cost must include separately reported in-kind support.

**Retained reentry:** proposed primary outcome, defined in the evaluation plan. **Status:** unmeasured. No effect size or success rate exists.

**Shortest path:** communication shorthand for a small useful evidence-supported recovery route. **Status:** heuristic goal, not global mathematical optimality. Public copy now uses the GPS metaphor without an optimal-path guarantee.

**Local persistence:** browser storage for route/answers. **Limit:** no account sync, no guaranteed offline initial load, no automatic school reporting, no encrypted learner account store.

## Design references

Khan Academy's rendered homepage informed readable surfaces and typography. The user subsequently selected green #14BF96, white #FFFFFF and navy #0A2A66. The original recovery engine supplied mathematical typesetting and route geometry. Nicky Case informed explanation through interaction. No third-party result or customer story is reproduced as evidence for Khanpanion.

## September 11 release and application refresh

The current code release includes original mathematical models, bounded recovery probes, a mascot introduction/spotlight tour, independent structural families, mixed rehearsal, spaced review and confirmed scope receipts. Verification is engineering evidence: 83 unit tests, 19 browser journeys and a 30-route build. No learner-effect size has been observed. See `RELEASE_2026_09_11.md` and `research/2026-09-11/LEARNING_DESIGN_DECISIONS.md`.

The seven application drafts in `submission.json` were rewritten for this release. Word counts stay below 300; the live application field rules remain authoritative. The earlier 300-character executive-field finding is identified as an earlier inspection rather than a current verified constraint.

Needs facts rechecked September 11: OECD, PISA 2022, 16% of assessed Philippine students at least Level 2 in mathematics: <https://gpseducation.oecd.org/CountryProfile?plotter=h5&primaryCountry=PHL&topic=PI&treshold=5>. PSA, 2024 NICTHS, 48.8% of households with internet access at home: <https://psa.gov.ph/content/percentage-households-internet-connection-increased-488-percent-2024-two-every-three>. These have different denominators and do not describe Khanpanion participants. Neither is presented as the latest result of all assessments or as a causal explanation for individual errors.

Budget assumptions preserved: PHP 100,000 total; allocations 24,000 + 18,000 + 18,000 + 16,000 + 8,000 + 8,000 + 3,000 + 5,000. Proposed 80-120 enrollees imply approximately PHP 833-1,250 per enrollee including setup, not cost per active learner or per learning outcome. Proposed pilot, school access, funding and outcomes remain unconfirmed except UP Manila endorsement supplied by the team.

The new slide plan must cite claims visibly in the submitted PDF and embed source hyperlinks. Presenter notes alone do not survive as evidence in a PDF. No competitor comparisons, private editor URLs or unverified efficacy claims belong in the presentation.

## September 13 submission opening

The current deck opens with 79% of Filipino 15-year-old students below baseline mathematics proficiency. The OECD's PISA 2025 Philippines country note, published 8 September 2026 and checked 13 September, reports 21% at or above Level 2. The opening uses the complement, 100% minus 21%. It describes school students represented by PISA and supplies national needs context; it is not a Khanpanion outcome or an explanation of any individual learner's difficulties.

Source: [https://www.oecd.org/en/publications/pisa-2025-results-volume-i-country-notes\\_2d4ff9ea-en/philippines\\_3699b9d5-en.html](https://www.oecd.org/en/publications/pisa-2025-results-volume-i-country-notes_2d4ff9ea-en/philippines_3699b9d5-en.html)

The September 11 budget and pilot figures above are historical. Current submission allocations and the public-launch model are in `output/submission/PROJECT_ANSWERS.txt` and `docs/BUSINESS_AND_SUSTAINABILITY.md`. No classroom access is required for the initial launch.